



UNIVERSITÀ
DEGLI STUDI
FIRENZE

Scuola di Ingegneria

Corso di Laurea in Ingegneria Informatica

Resiliency Real-Time and Certification

Safety AI

Edoardo Sarri

7173337

Giugno 2026

Indice

1	Introduction	5
1.1	Tool used	5
1.2	Terms	5
2	Litterature	7
2.1	Type Of AI	7
2.2	Automation Level	7
2.3	Safety Standards	8
2.4	Point of Application	8
3	AI-Based Safety-Critical System	10
3.1	Product	10
3.1.1	Testing	10
3.1.2	Neural Network	10
3.2	Runtime	11
3.3	Process	12

Elenco delle figure

Listings

1 Introduction

In the last years an interesting challenge in the field of Artificial Intelligence (AI) and safety-critical system is to integrate the AI model in those systems that require a predictable and safety behaviours. The black-box nature of the AI models implies the absence of structured standard and methodologies to obtain AI-based safety-critical system.

In this project i will start the study on these topics from one of the main paper [1] on this discipline.

1.1 Tool used

During the development of this project, in order to improve the report, the development and the quality of the whole work, i used the following tools:

- **NotebookLM**

It was useful to perform a first review of the guide line paper [1]. [Here](#) there's the link of the notebook: despite the chat history aren't longer available, you can see the main interaction like concept map and report.

- **Copilot completions**

During the write of this report (and of the slides), the Copilot completions generate for me some part of the text.

- **Gemini CLI**

The CLI version of Gemini was used to check the grammar correctness of my english after the writing.

1.2 Terms

In this little Section we make some clarification about the terms used in the following of the work.

- **AI and Machine Learning**

This refers to the algorithms techniques that allow us to create models that learn patterns from the data.

- **AI Safety**

This is used to refer to the techniques used to avoid or mitigate the potential catastrophic harm of the AI systems to humans. This means that with AI safety

we can also refer to the methodologies that allow us to deploy AI systems in safety-critical system, being compliant with the safety standards.

- **Functional Safety (FuSa)**

The Functional Safety is defined as "part of the overall safety of a system that assures the freedom from unacceptable risk" [120] [1]. The aim of FuSa is to ensure that a system is free from unacceptable risk.

2 Litterature

In this Cahpter we see some classification related to the AI safety-critical system. This is useful to have a global of the context in witch we will moving after.

2.1 Type Of AI

In the litterature there're huge Type Of AI (TAI). In this Section we made a little recap on them: maybe this will be useful in the following.

- **Connectionists**

These are models where the training is performed by an optimization technique, where the most used is the gradient descent. Are example the Neural Networks (e.g., CNN, Transformer).

- **Bayesian**

The result is obtained in a probabilistic way.

- **Symbolists**

They use the logic to perform the reasoning (e.g., decision tree and Prolog). Given that the inference must validate all part fo the logic (i.e., each component of a boolean expression) learned in the learning phase, these models are usefull if we want have a explainable model (e.g., in a **prediction of the fault** in real-time scenario).

- **Analogizers**

Usually these are classification model based on a similarity score (e.g., KNN, SVM).

2.2 Automation Level

In the field of AI we can define several level of automation. A classification can be made, from the highest to the lowest, in the following way:

1. Autonomous systems operate in an open environment.

The ISO 22989 defines that this is the highest level of automation, in which the system can modify its behavoir and goals without the human control.

2. Heteronomous system operate in a (semi-)open environment.

This include different level of automation where the requirements can be defined in a range but they aren't well known and the human control can be variable.

3. Automatic systems operate in a closed environment.

In this case the system execute in a given environment we the requirements are known in the design phase.

4. Collaborative Robots

There are robots the work side by side with humans without having total independence.

2.3 Safety Standards

Moving from the traditional way to guarantee the safety in a system to using the AI to achieve this, we must define also a new class of standard. The most common problem is related to the probability of failure: the most critical level of SIL (or ASIL or DAL) requires a very low probability of fails (i.e., 10^{-9}) and an excellent accuracy for a AI model is 99%, the common AI models aren't compliant with these requests.

- **Traditional safety standards**

Their objective is mitigate the risk using programmable function, the Safety Instrumental System (SIS), that are embedded in the system. The quality of the SIS are defined by different measures depending of the domain: we have the Safety Integrity Level (SIL) for the industry (IEC 61508), the Automotive Safety Integrity Level (ASIL) for the automotive (ISO 26262) and the Design Assurance Level (DAL) for the aerospace (DO-178C).

- **AI Safety Standards**

After have defined the Heteronomous system (e.g., Advanced Driver Assistance Systems (ADAS)), we can say that we can have a failure also in the absence of a hardware or software fault. This means that these new algorithms must be compliant to the Safety Of The Intended Functionality (SOTIF) standard (ISO 21448), that bring them to be compliant also with the classical FuSa.

2.4 Point of Application

In this Section we made a recap on the kind of different application of the AI: there're three categories that can describe the *when* and *on what* we can use a AI model.

- **Product**

A safety-critical system relies on one or more safety functions that are based on AI model. This model is offline: the training is performed before its embedding in the system.

- **Runtime**

It's similar to the previous one, but the model in this case can update its self during the execution.

- **Process**

The system isn't provide by any AI model, but the system is designed and developed using some AI model (e.g., generative AI to generate code or perform the tests).

Based on this classification, the ISO 5469 show us three classes that define when AI models can be include in a safety-critical system (i.e., product, runtime and process) remaining complied with the standard defined in Section 2.3.

- **Class I**

This systems are based on the tradition AI model, like those we have referred as *Symbolists* in Section 2.1. Since they're totally deterministic and we can perform formal verification, these models are complied with the classic FuSa.

- **Class II**

The system have inside it one AI model that can be also non complied with the FuSa. The compliance is given by the application of a run-time checker, usually known as *safety bag*; the idea is that if the model fails, the checker notice it and takes a safe choice.

- **Class III**

These are the systems that can't include inside them one AI model. In this case usally the solution is include the model, but the engineers doen't take the reponsability of possible failures (e.g., ADAS in autoveicle).

3 AI-Based Safety-Critical System

Although inserting AI in a safety-critical system can be very challenging, there are a lot of techniques that can be used and that are already tested. For example, referring to the classification in Section 2.4, we have that the *class I* and *class II* contain systems that already are on the market (e.g., SIL4 railway interlocking).

3.1 Product

As said in Section 2.4 the product class describes a safety-critical system that relies its SIS on one or more AI-based functions trained offline. This function can be "simple", like in the case of a symbolist AI model, and so belongs to the *class I*, or can be very complex and so belongs to *class II*: in the first case, since the model is deterministic, we can use a formal verification to let the system comply with the FuSa; in the second one the standard industry technique is to use a runtime checker that takes a safe choice on the next action if the model operates outside the requirements or produces an unsafe result.

3.1.1 Testing

The main problem of these AI-based safety-critical systems is the assurance on the safety that we are able to assign them. This derives from the stochastic nature of the AI models, which implies that we can have some *uncontrollable failure*; to reduce this uncontrollable we should search among something that is unknown. For this, when we describe the safety function in this scenario we speak about intended functionality: their design are a set of high-level objectives rather than a set of precise and well defined instructions. This can be mitigated by applying some formal method verification, but also in this case the black-box nature says us that this method cannot be the only one. Another problem is the testing of the AI model, that is an area very explored (e.g., testing generation from a NPL generation model).

3.1.2 Neural Network

Although the symbolist models provide understandability and explainability of their decisions the most powerful models are the Neural Network (NN).

When we are dealing with this kind of model, we have to go into detail with some problems:

- The supervised learning relies on data. When we have to manage (i.e., pre processing and the data division in train and test set) the data we have to do this with

very high accuracy.

- This is the area where *change nothing change everthings* [1]. This means that changing activation functions and hyper parameters implies having a very different behaviours.
- The NN training is iterative; this must be included in the V-Model within specification and testing phase.
- The AI framework (e.g., Pytorch) are general purpose and friendly-oriented. There're instead some APIs that provide also a set of safety requirements.

3.2 Runtime

As discussed in Section 2.4, a runtime safety-critical system is able to change its AI module during the system execution. These are notable system because allow the SIS to evolve with the context: a case study report the gas oil turbine where the safety function evolve with the engine degradation.

This kind of solution based its dependability and the standard compliance on four difference techniques:

- **Safety Bag**

This is the same technique saw above: a runtime checker monitoring the AI model result and if it see a strage behavior it provide a safe result.

- **Safe Adaptation**

Both the AI model and the (continous) training algorithm are conformed to the safety standards.

- **Limited Adaptation**

The (continous) training is limited within a safe range: this means that also if the weiths of the model change, they cannot do with high magnitude.

- **Limited Actuation**

In this case the adaptation is performed with some hard costraints in the result of the resulting model: for example the new model cannot have a output where the magnitude of result is grather than a threshold.

- **Library-Based Offline**

We train a set on AI models where each of them is conformed to the safety standard; during the inference, the system choice among all possible models in a non linear way (i.e., with if/else).

3.3 Process

These are offline solution the help and support the engineer to craft the safety-critical systems, both from the point of view of the whole system and the AI item that implement the safety function.

In the traditional safety engineering, the AI model can be usefull the validate and assess the V-Model: during the specification we can use a NLP model (e.g., transformers-bases) to discover the hazard and the issue in the process; during the design and the implementation the model can be useful to generate code; during the testing phase they become usefull to generate test case.

If we look to the use of AI to costruct other models that will implement the safety function, we have different solution: in the AutoML we use the reinformcement learning to found the best model hyperparameter; we can implement model that search in a exhaustive way all combination for test our system.

Bibliografia

- [1] Jon Perez-Cerrolaza et al. «Artificial intelligence for safety-critical systems in industrial and transportation domains: A survey». In: *ACM Computing Surveys* 56.7 (2024), pp. 1–40.